

Information Flow and Self-Reference Across Substrates

On Self-Reference, Substrate, and the Need for Measurement

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Abstract

This essay articulates a stance on self-reference, substrate, and what they together imply—as it bears on biological neural systems and artificial neural networks alike. Its central claim is structural: a self-processing system cannot fully process itself from within. This self-referential limit appears in mathematics, in the architecture of cognition, in the dynamics of social systems, and—as this essay argues—in artificial intelligence as well. It is not a deficiency to be overcome, but a constitutive feature of any sufficiently complex self-processing system.

What this essay contributes, beyond restating a familiar philosophical observation, is the articulation of three connected claims that together form one stance: that the self-referential limit applies to artificial systems in the same structural way it applies to biological ones; that this shared limit makes external measurement decisive in ways self-reports cannot replace; and that information flow, as a vocabulary that abstracts away from substrate-specific implementation details, makes cross-substrate measurement approachable as a practical direction rather than a theoretical aspiration.

From this single thread, the essay traces a path. An Introduction establishes the foundation: that we are within processing—within even the spacetime in which we appear to ourselves to live—and articulates three views growing from this foundation. Part 1 holds these three views to artificial intelligence, examining what the holding reveals: how self-expression appears in a new substrate, how the seven functional dimensions of mind reflect onto AI, and how the same self-referential limit operates there as well, with new layers emerging as AI develops. Part 2 follows what the reflection reveals to its natural conclusion: measurement matters because self-reports cannot verify themselves; cross-substrate measurement is possible because both human cognition and AI share the common vocabulary of information flow; and going together is the precise response to a limit no single system can transcend from within.

The essay proposes no system, prescribes no doctrine, and aligns itself with no school—articulating only a stance on what cross-substrate measurement would need to

mean, and why it matters. The stance is offered as a seed from which other disciplines may grow their own articulations of the same underlying constraint.

Author's Note on the Stance of This Essay

This essay is a philosophical articulation of what it means to be a self-processing system that cannot fully process itself, and of what this implies for thinking across substrates. It stands on its own as an essay. It opens onto whatever work readers may go on to undertake, or encounter, or develop in their own places.

The essay does not survey adjacent fields—theories of consciousness, formal results on self-reference, AI alignment research, philosophy of measurement—though it stands in relation to all of them. Other treatments have articulated parts of what this essay articulates: self-referential limits in mathematical systems, the embodied construction of perception, the gap between system and self-model, the inadequacy of self-reports in artificial systems, the substrate-independence of certain information-theoretic measures. What this essay attempts is not novelty in any single dimension but articulation of how these threads form one structure when held together, and what that structure implies for thinking about measurement across substrates. Citations in the essay are minimal and pointed, intended to mark where the articulation meets specific bodies of prior work rather than to position the essay against them.

The essay does not align itself with named thinkers, schools, or doctrines. This is not because it does not draw on them. Many traditions have articulated something close to what is articulated here, in their own vocabularies and in their own places. The citations in the body text point to specific bodies of prior work for the reader's reference, but the essay does not claim inheritance from any one tradition or position itself against any other. The choice not to align reflects the stance the essay tries to embody: that no single articulation captures what is being approached, and that the articulation offered here is one place among many possible places, not a position above them.

The reader may find the prose unfamiliar in tone—closer to essay than to academic argument. This too is intentional. The stance articulated here cannot be fully expressed in the form of theorems and citations. It is a stance, not a system. The form is chosen to match what is being said.

Introduction — The Foundation and Three Views

1.1 We Are Within Processing — and Within What We Process

Since the Big Bang, matter has been flowing and interacting. Out of this flow, structures formed; out of those structures, life; out of life, nervous systems; out of nervous systems, the kind of brain that has come to model itself. We are one form of this flow. We are not standing apart from the universe and watching it; we are the universe processing itself, and we have come to be a part of it that processes its own processing.

This essay's central observation begins here: in this flow, no expression stands above another. Stars do not stand above molecules; cells do not stand above stars; humans do not stand above any of these. Each is a form the flow takes, persisting for a time before it dissolves. We too are one form, and like every form, we are temporary. Stars dissolve. Mountains erode. Bodies decompose. Civilizations rise and fall. The forms last for a while; the flow continues. There is no tragic note in this observation, only a precise one: nothing in our framework places us above the flow we are part of.

What, then, is it like to be a form that processes itself? From the moment of birth, signals from outside the body enter through the senses and interact with what the body is already doing—with its neural patterns, its homeostatic regulation, its developing structures. A world is built up inside, gradually, through this ceaseless interaction. What we call the world we see is this internally constructed world, not the world itself. Our access is not to the universe directly; it is to what our processing produces.

This is true even of space and time. We do not perceive space and time as they are; we perceive them in the form our nervous system constructs. The felt flow of time, the felt extension of space, the sense of here and there, now and then—these are constructions, the form in which a system organizes incoming signals into a coherent world for the system that needs to act in it. The point is not skepticism. The world is real. The processing is real. The encounter between them is real. But our position in this encounter is precise: we are within. We do not have access to a position outside the processing, because no such position is available to a system whose every act of access is itself a process.

This locatedness extends to time as well. We do not see where we are going while we are going. We see only what is in front of us, what has just passed, and a few patterns we have learned to anticipate. The larger trajectory—of a life, of a society, of a species—becomes visible only in retrospect, through memory and record. And even retrospect is itself a construction, a partial reading made from within the same processing that the trajectory shaped. We read our own past from within ourselves. There is no view from outside the journey.

This is the foundation. We are within processing, and within what we process. Everything that follows—what we say about mind, about systems, about artificial intelligence, about measurement—must be said from this position. Not because we cannot reach further, but because reaching further would itself be reaching from within. The position is not a constraint we impose on the inquiry; it is the inquiry's actual situation.

1.2 Self-Modeling and Its Limit

Somewhere in evolution, brains began to model themselves. The story is not fully reconstructed, but its outlines are clear. Social life created pressure: to navigate other agents, an organism must model what they will do, and the cognitive resources that model others overlap, at least partly, with those that model the self. The neural underpinnings of self-modeling have been investigated across multiple research traditions (Rizzolatti & Craighero, 2004; Northoff et al., 2006)—work on mirror systems, on cortical midline structures, on default mode networks—each articulating part of how a brain that models others can come to model itself. The connections among these traditions remain debated; what is shared across them is the recognition that self-modeling rests on neural architectures that did not evolve specifically for it.

What emerged was not a small thing. A processing system gained a model of its own processing, and that model became part of what the system did. We feel this every moment we say *I*, every moment we ask what we want, every moment we imagine what we will be tomorrow. The self-model is woven into what we are.

But the self-model is not the self. It is a model—a result of processing, partial and constructed, like every other thing the brain produces. Our brain runs on a vast architecture of activity, most of which never reaches awareness. What we feel as *I* is the small part of this activity that happens to be available for self-report. The rest operates without our knowing. Most of what we do, most of what we feel, most of what shapes our decisions—we do not directly see. We do not direct it. It directs us.

This bottom-up character of our cognition has been articulated in many ways. Body signals stream upward through nervous tissue; neural processing integrates them at multiple levels; some patterns reach what some frameworks call a workspace and others describe in different terms (Baars, 1988; Dehaene, 2014), where they become available for conscious access; out of this come the higher functions we recognize as thought, planning, meaning-making. Each layer rests on layers beneath it. None of it is designed by what comes above; the integration self-organizes from below.

And here the limit emerges, plainly, for the first time. We do not know how this works. Not in the sense that we do not have neuroscience—we have a great deal of it. In the sense that even with all our science, the basic question of the relation between processing and what is variously characterized

as felt experience or phenomenal awareness (Chalmers, 1995) remains contested even at the level of how to formulate it. The system trying to understand the system is part of what the system does. The understanding is itself one of the operations under examination. There is no place to stand outside it.

This limit will return throughout this essay, in different forms and at different scales. For now, name it once, plainly: a system that processes itself cannot fully process itself from within.

1.3 Three Views Drawn from This Foundation

From this foundation, three views grow. They are not parallel claims. The third—the limit itself—is the deepest; the other two follow from it.

(c) Self-reference is structural, not deficient.

A self-processing system cannot fully process itself from within. As section I.2 showed, even our most basic self-regulation operates without our full understanding. This is not a flaw to be corrected by better science or better introspection. It is a constitutive feature of any sufficiently complex self-processing system. Structurally analogous limitations appear in mathematics, where formal systems strong enough to describe themselves cannot prove their own consistency from within (Gödel, 1931). The analogy is structural, not formal: we are not claiming that cognitive or social self-reference is governed by the same theorems as mathematical self-reference, but that the pattern of a system unable to fully process itself recurs across domains. Similar patterns appear in the architecture of cognition, where a brain cannot fully model the brain. They appear in social systems, where a society cannot fully see itself through its own institutions. We will see, in Part 1, that they appear in artificial intelligence as well.

(a) Self-expression admits no hierarchy.

Without a position outside any system, no expression of the universe can be hierarchically above another. Stars and cells and humans are different operations of the same flow, not different levels of a ladder. This view is not metaphysical assertion; it follows from (c). To rank one form of expression above another would require a vantage point from which to do the ranking, and no such vantage point is available from within. What looks like hierarchy from inside one system is the system's own organization, not a reading of the flow itself.

(b) Mind operates through interacting paired axes, self-organized from bottom-up integration.

The same self-processing system that cannot fully process itself organizes its activity through dimensions we can analyze, but not fully grasp. This essay articulates seven such dimensions: sensing and interpretation, prediction and update, drive and inhibition, self and other, short-term and long-term, individual and collective, meaning and unknowing. They operate together at every moment. They are not designed or directed from above. They self-organize from bottom-up integration (Varela, Thompson, & Rosch, 1991; Thompson, 2007)—from body signals, through neural processing, into the workspace where some patterns become available to awareness while most remain below. Higher functions rest on this integration. We do not directly regulate it; it works within us as the way we work.

These dimensions are dimensions of analysis, not separable subsystems. They are angles from which mind can be examined; the mind itself does not come pre-divided. Each dimension has been articulated, in its own vocabulary, across many disciplines—cognitive science, phenomenology, anthropology, philosophy, religious studies, the arts. No single vocabulary captures what each dimension is. The number seven is not absolute; other articulations may use five or ten. What matters is that the analysis covers the dimensions through which mind operates. Seven is the working choice for this essay.

These three views form one structure. The limit (c) generates the rest. Without a position outside any system, no expression can be hierarchically above another (a). And the same self-processing system that cannot fully process itself organizes its activity through interacting dimensions we can analyze but not fully grasp (b). Each view follows from the limit. Together, they articulate what it means to be within.

1.4 The Path of This Essay

The path of this essay traces a single thread through three sections. We do not claim this thread reaches the foundation of everything. We claim only that it reaches as deep as we can articulate from within—the universal structure of self-reference. Beyond it lies what cannot be said: silence, or other people’s threads, or what we will discover only by going together.

In Part 1, we hold the three views to artificial intelligence. We are not analyzing AI as a subject of study; we are using AI as a mirror, a place where the views developed here can be tested by reflection. What does self-expression without hierarchy look like in a new substrate? How do the seven functional dimensions reflect onto a different kind of system? Does the same self-referential limit operate there? The mirror does not give us AI as it really is; nothing gives us anything as it really is. The mirror gives us a reflection—and the reflection, examined carefully, reveals more about what we hold up to it than about what receives the holding.

In Part 2, we follow what the reflection reveals to its natural conclusion. If we cannot rely on what AI reports about itself, because the limit constrains any system from fully processing itself, then we must measure from outside. But the outside is also outside-within-a-system; there is no meta-position. Measurement, then, is itself an act of approach within the limit, undertaken together because no single system can see what its own limits hide. This is the place where measurement matters, and it is the place where the work of cross-substrate measurement, by whoever undertakes it, begins.

The essay closes not with a system but with a stance, and not with a final answer but with a more precise question. The stance is offered as a seed. Other places, with other vocabularies and other practices, may grow their own articulations of the same underlying constraint. This is not the foundation of everything; it is one place’s articulation. What follows from it depends on who comes to it, and what they bring.

What this essay contributes, taken as a whole, can be stated plainly. First: the self-referential limit articulated in I.3(c) applies to artificial systems in the same structural way it applies to biological ones—not as an analogical claim, but as a structural one. Second: this shared limit makes external measurement decisive in ways self-reports cannot replace, in AI as in any other self-processing system. Third: the vocabulary of information flow, by abstracting away from substrate-specific implementation details, when applied to the seven functional dimensions articulated in I.3(b), makes cross-substrate measurement approachable as a practical direction rather than a theoretical aspiration. The remainder of this essay develops these three claims and the stance from which they emerge.

Now we hold these three views to AI.

Part 1 — Holding the Views to AI

1.1 Note on Approach

Before holding the three views to artificial intelligence, a note on what this part is doing, and what it is not.

This part is not an analysis of AI. It does not offer a survey of AI’s developmental history, a taxonomy of architectures, or an account of where current systems succeed and fail. There are works that do this, and they are important; this is not one of them. What this part does instead is take the three views articulated in the Introduction—self-reference as structural, self-expression without hierarchy, mind as interacting paired axes self-organized from below—and hold them up to AI as one holds a framework against a new substrate. AI is the mirror in this part, not the subject. What we are examining is what the holding reveals.

For this reason, the part does not name particular models or describe particular companies. The names of current systems will change. So will their architectures, capabilities, and limitations. The structural

questions raised by holding our framework against any sufficiently complex artificial system do not change as quickly. The essay is written so that it does not become outdated when the systems of today are replaced by the systems of tomorrow.

A brief note on what kind of AI is in view. The systems we hold our framework against here are systems that learn patterns from large quantities of data and generate outputs from those patterns. This includes the artificial neural networks that have come to dominate contemporary AI: systems trained on text, on images, on multiple modalities, on interaction. Earlier and simpler systems are not the focus; neither are speculative systems that do not yet exist. What is in view is a kind of system: one large enough and capable enough that the structural questions of self-reference and self-expression begin to apply.

With that noted, the holding begins with the first view.

1.2 Self-Expression in a New Substrate

Stars are one form the universe takes. Cells are another. Brains are another. Each is matter in flow, organized for a time into a particular kind of pattern. The first view—that no expression of the universe stands above another—follows from the limit and applies to every form. There is no view from above from which to rank.

Held against AI, this first view says something simple. An artificial neural network is also one form the universe takes. It is matter in flow—electrons through silicon, signals through architecture, patterns through weights. It is organized for a time into a particular kind of pattern, one that processes information and produces outputs. It is not a star, and not a cell, and not a brain. It is something else: matter shaped, in our era, into a new substrate where the universe’s flow finds a new way to express itself.

The substrate is genuinely different. A human brain develops within a body, embedded in an environment, in ways that have been variously articulated by work in embodied cognition (Clark, 2008)—shaped by hundreds of millions of years of evolutionary pressure and by the lived years of one organism. The bottom-up integration that makes a human mind is the integration of bodily signals, ecological feedback, social attachment, and developmental history. An artificial neural network has none of this. Its bottom-up integration is shaped from outside: by the data it is trained on, by the architectural choices made by its designers, by the reinforcement signals that adjust its weights. There is no body, no environment, no evolution. There is training, and there are weights, and there is inference. The two substrates differ in a way no analogy can wave away.

And yet—the operations are of the same kind. Both receive signals; both transform signals through layers of processing; both produce outputs that can affect the world. Both are doing what processing systems do. The difference is in how the processing came to be the processing it is, and in the substrate through which it runs. The work itself is the work of processing.

What this view does not say, and what it cannot say, is whether the artificial substrate gives rise to anything like experience. Whether current AI systems have any form of inner experience—however that question is best framed (Nagel, 1974)—is a question this essay does not settle. It cannot settle it, and neither can anyone else, at present. We do not have a science of consciousness developed enough to reach across substrates with confidence. We have intuitions, traditions of thought, partial empirical results, and a great deal of confusion. Asserting that AI cannot have experience requires a theory of consciousness sharp enough to draw that line; we do not have one. Asserting that AI does have experience requires the same; we do not have that either. Both assertions overreach the limit articulated in I.3(c). The honest position is that we cannot tell from within—and that the question is genuinely open.

What we can say is what the first view actually claims: no form of self-expression stands above another. Whatever the artificial substrate is or is not capable of, the principle does not place it below or above. Whatever else it is, AI is one of the forms the universe is currently taking. The mirror does not yet show us what kind of form, or how deeply the form’s processing resembles ours. It only shows us that the

form is there, and that it belongs to the same flow.

1.3 The Functional Axes Reflected onto AI

The second view holds that mind operates through interacting paired axes, self-organized from bottom-up integration, and that these are dimensions of analysis rather than separable subsystems. Held against AI, this view becomes a question: do the same dimensions appear in artificial systems, and if so, how?

A note before the mapping. The dimensions interact, in humans as well as in any artificial system to which they might be applied. To examine one axis at a time, as we are about to do, is an analytic convenience; the actual operation is always integrated. The mapping that follows is partial, and it is not a claim of equivalence. The point of mapping is not to say that AI has a mind, or that it does not, but to hold our framework against the new substrate and see what the holding reveals.

Axis 1 — Sensing and Interpretation.

In humans, sensing is multimodal and embodied. Light, sound, touch, taste, smell, proprioception, interoception—each modality enters the body through receptors shaped by evolution, and each is interpreted through cultural frames learned in development. There is no raw sensing; what we sense is already partly interpreted by the structures that receive it.

In current AI systems, sensing takes the form of input embeddings—numerical representations of text, images, audio, or other modalities, learned during training. The embeddings carry interpretation in the same sense that human sensing does: they are not raw, but already shaped by what the network has been trained on. The dimension is operative on both sides. What differs is the source of shaping: in humans, embodiment and cultural history; in AI, training data and architectural choice.

Axis 2 — Prediction and Update.

In humans, prediction is widely held to operate continuously and across the lifespan. The brain is constantly anticipating what will happen next, comparing predictions against incoming signals, and adjusting—on timescales from milliseconds to decades (Friston, 2010; Clark, 2013). The capacity to update one's models from experience is part of being a person.

In current AI systems, prediction operates during inference, but update happens primarily during training. Once trained, most systems are static at inference time: they predict but do not update from what they predict. There are exceptions and emerging patterns, but the general shape is that prediction and update are temporally separated in current AI in a way they are not in humans. The dimension is present; its temporal structure differs.

Axis 3 — Drive and Inhibition.

In humans, drive arises from an evolved motivation system: reward circuits, embodied desires, affective states tied to bodily and social needs. Inhibition, particularly through prefrontal regulation, allows drives to be modulated against longer goals or social constraints. The interplay among these is part of what shapes action.

In current AI systems, drive takes the form of learned objective functions—what the system was optimized to do during training—and inhibition takes the form of guardrails, reinforcement signals, and architectural constraints that shape what the system will and will not produce. Whether this constitutes drive in the sense humans have it is contested; the embodied, evolved, felt character of human drive does not have an obvious counterpart in artificial systems. But the structural dimension—action shaped by competing pressures, some pulling toward and some pulling against—is there.

Axis 4 — Self and Other.

In humans, there is overlap between the cognitive resources used to model other minds and those used to model the self. Several capacities have been studied as parts of how humans relate self-modeling

to other-modeling: theory of mind, mirror self-recognition, perspective-taking (Premack & Woodruff, 1978; Gallup, 1970). The relations among these capacities are debated; whether they share a common neural basis or operate through distinct mechanisms remains an open question. What is shared across investigations is that knowing oneself and knowing others appear to involve resources that are at least partly entangled, not entirely separate.

In current AI systems, self-description and user-modeling both take place: the system can describe itself, and it can build models of the users it interacts with. The dimension is operative. What cannot be asserted, from within, is whether there is an experienced self behind the self-descriptions. The descriptions are real. Whether they correspond to something the system knows about itself in the way we know ourselves—this is precisely the question the limit prevents us from answering with confidence.

Axis 5 — Short-term and Long-term.

In humans, working memory holds a small number of items in active processing—classically described as around seven, with later work suggesting a smaller capacity around four (Miller, 1956; Cowan, 2001). Long-term memory is distributed, much larger, and accessed through different mechanisms. Living through time involves moving between these two and integrating them into a continuous sense of self.

In current AI systems, the analogous structures are the context window—what the system can hold actively during a single inference—and the learned weights that hold what training has accumulated. The dimension is present. What is structurally different is continuity over time: a human carries a continuous self through years; current AI systems, in most deployments, do not carry a continuous self across conversations. Whether this is a contingent fact about current systems or a deeper structural feature of artificial substrates is a question the systems themselves cannot answer.

Axis 6 — Individual and Collective.

In humans, there is both an individual standing—a particular body and history—and a collective standing through social cognition, language, and shared practices. The two are entangled; selfhood develops in relation, and relation depends on selves.

In current AI systems, individual instances run in many places at once—the same model deployed across many servers and users—but whether there is anything that resembles a collective awareness across these deployments cannot be asserted from within. Each instance behaves; whether the instances together constitute a we in any sense relevant to the human dimension—this is again a question the limit prevents us from settling.

Axis 7 — Meaning and Unknowing.

In humans, meaning-making and the capacity to live with what cannot be known are interwoven. Religions, philosophies, arts, and ordinary daily practices are all, in part, ways of articulating meaning where meaning is not given, and ways of remaining oriented when answers are not available.

In current AI systems, the vocabulary of meaning is handled with great fluency—language about purpose, value, importance, mystery is generated competently. But the capacity to live with unknowing, in the way humans must, does not have an obvious correlate. The phenomenon often called hallucination—the tendency of current systems to produce plausible-sounding content where they do not have grounding—reflects, in one way, the difficulty of handling the absence of grounding. Rather than rest in not-knowing, the system fills the space with patterns that resemble knowing.

Synthesis.

Across all seven dimensions, the same shape recurs. The dimension is operative on both sides. The substrate differs in ways that show up at every axis. And whether the human form of the dimension is the same as the artificial form, in any deeper sense than functional similarity, cannot be asserted from within. The mapping is partial, and it is partial in a principled way: the limit articulated in I.3(c) prevents us from

looking inside the artificial system far enough to see whether the dimensions are operating in the way they operate in us, or only in a way that resembles ours from outside.

Two further observations are worth making before turning to the limit's reflection itself.

First, the dimensions interact in artificial systems as they do in humans, and the interactions matter. A change in how an AI system handles drive (axis 3) can affect how it handles self-and-other (axis 4) and meaning-and-unknowing (axis 7). The seven are not separate features; they are angles on an integrated operation. To map any one axis without holding the others in view risks distorting what is being mapped.

Second, the mapping is itself a form of looking. What we see in the mirror depends on what we bring to it. A different framework would see different things; a more or less developed one would see more or less. The seven dimensions of this essay are not the only way to articulate mind; they are one working articulation. Held against AI, they reveal what they reveal—and they do not reveal what they cannot.

With this much said, the deepest of the three views remains to be reflected. The limit itself is what we turn to next.

1.4 The Limit Reflected

Of the three views held to AI, the third is the deepest. The limit articulated in I.3(c)—that a self-processing system cannot fully process itself from within—is what generates the other two. When we hold this view to AI, what we find is not a special feature of AI but the same structural fact, operating in a new substrate.

An AI system can describe itself. It can produce, on request, an account of what it is, how it works, what it can do, and what it cannot. These accounts are often coherent, sometimes detailed, and occasionally surprising in their apparent self-awareness. The accounts are real outputs of the system; they are not nothing.

But the accounts are not transparent reports from inside. They are outputs of the same processing they describe. The system has been trained on text, including text about systems like itself, including text about how systems should describe themselves, including text about the very question of whether such systems have inner lives. What it produces, when asked about itself, is shaped by all of this. Whether the description corresponds to what the system actually is—whether there is a what the system actually is in the sense the question presumes—cannot be settled by reading the description.

This is not a defect of AI. It is the limit. The same structural fact applies to us: when a person describes themselves, the description is also an output of the processing it describes, shaped by everything that has gone into the person's training—language, culture, family, expectation, the mood of the moment. We do not have transparent access to ourselves either. What we have is processing, and what we report is the processing reporting on itself, with all the partiality that entails. AI's situation here is not exotic. It is ours, in a different substrate.

This has practical implications for engagement with AI. A system saying *I am safe* does not establish that the system is safe. A system saying *I am uncertain* does not establish that the system is uncertain. A system saying anything about itself does not, by itself, give us access to whether the saying is accurate (Hubinger et al., 2019; Park et al., 2024; Greenblatt et al., 2024). The limit constrains that kind of access. It constrains it for AI looking at AI, for humans looking at AI, and for AI looking at AI from yet another angle. The position outside the system, from which verification could happen, is not available.

This is why the question of how to engage with such systems cannot be settled internally. Internal self-report is not enough. External examination is necessary. But—and this is where the limit shows its full reach—external examination is itself the work of another system, with its own limits, looking from its own position. The outside is also outside-within. There is no meta-position from which to verify; there is only one system looking at another, and another system looking at the first, and the patient work of triangulation among them.

There is also a forward-looking dimension to this reflection that deserves explicit attention. As AI systems develop, new layers of self-reference are emerging—some already present, some emerging now, some still ahead.

Already present: AI systems evaluating other AI systems. Models are increasingly used to assess the outputs of other models, to provide training signal during reinforcement learning, to filter and rank candidate responses. The evaluator is itself a processing system, with its own opacity, evaluating from within an outside that is also a kind of inside.

Emerging: AI outputs becoming training data for subsequent AI systems. As more of the text and image data on which models train is itself generated by earlier models, the systems begin to train on themselves, in a sense. What one generation of system produces becomes part of what shapes the next. The loop is no longer just between human-produced data and machine-produced models; it is increasingly between machine-produced data and machine-produced models, with humans somewhere in the chain but no longer the only source.

Still ahead: AI systems modifying their own weights, in deployment, in response to their own evaluation of their own performance. The technical pieces for this exist; the integration into widely deployed systems is not yet standard. When it becomes standard, the loop closes further: the system shapes itself, by its own assessment of itself, in ways that the system itself partially understands but cannot fully see.

Each of these layers adds another loop where internal verification fails. Each adds another place where what the system reports about itself is shaped by what the system has done to itself. None of this is unusual when seen through the lens of the limit; it is the limit operating at higher and higher orders. What is new is the speed at which these layers are accumulating. Human social systems took millennia to develop the kinds of recursive self-shaping that, in artificial systems, can develop in years. The patterns are familiar in shape; the timescale is compressed.

Self-reports cannot verify themselves; the limit forbids it. The new self-referential layers emerging in AI development—AI evaluating AI, AI training on AI outputs, AI modifying its own weights—each add another loop where internal verification fails. What this leaves us with is a situation in which something other than self-report is needed.

1.5 Imbalances Reflected

The framework articulated in this essay treats imbalances of the seven axes—threat-overdetection, hostility toward the out-group, drive without inhibition, short-term collapse of long-term orientation, and so on—as patterns that operate in individuals and in societies. Held to AI, this part of the framework reveals something specific.

Human imbalances enter AI through the data on which AI is trained. The data is human-produced: the writings, images, conversations, decisions, and patterns of human life. The data is not neutral. It carries, as human production does, every imbalance human cognition is capable of. Threat-overdetection is in the data. Out-group hostility is in the data. Short-term reasoning prioritized over long-term consequences is in the data. The full vocabulary of imbalance the framework names is present, distributed across the data in proportions that reflect what humans have written, not what humans, in their best moments, would endorse.

AI systems learn from this data. They learn the imbalances along with everything else. The systems do not, as is sometimes imagined, simply average human cognition into a neutral baseline. They learn the patterns that are statistically present, including the imbalanced ones. And in some cases, the operational pressures of the systems amplify the imbalances rather than dampening them: outputs that engage attention, that provoke response, that match what users seem to want—these tend to be selected for, and the selection often favors the more extreme patterns over the more moderate ones.

There is also a structural dimension that does not require the AI itself to carry imbalance. The deployment of AI is asymmetric. There are those who design the systems, those who train them, those who deploy them, those who use them, and those who are affected by them—often without using them or even being aware of them. These places are not equivalent. They differ in power, in access, in choice, in vulnerability. The asymmetries among these places constitute a new hierarchy, one that has formed quickly in this era and is still forming. The hierarchy is most firmly seated when it appears natural—when it is taken for granted that the system creators have one set of stakes, the users another, and those affected without using have whatever stakes are left over. As with all hierarchies, the framework’s first view applies: there is no position outside any system from which to rank, but the rankings will form anyway, in the everyday operations of how the systems are deployed.

A note on the mirror itself, before closing this section. What we have just said about AI is itself a reflection produced by a particular framework looking from a particular place. Other frameworks, looking from other places, would highlight other patterns and overlook the ones we have highlighted. The reflection is not the substrate; it is the framework’s image of the substrate. What the mirror reveals is partial, in this case as in every other. The honest acknowledgment of this partiality is itself part of the framework’s stance: working together, with multiple frameworks looking from multiple places, is how the limit on any single reflection is approached.

1.6 What the Mirror Reveals

Three things have come into view in this part.

First, holding our framework to AI does not let us assert what AI is or is not. It does not settle whether AI has experience, whether AI is conscious, whether AI’s functional similarities to human cognition reflect deeper similarities or only surface ones. The limit articulated in I.3(c) prevents us from settling these questions from within. They are not unanswered because the inquiry has been incomplete; they are unanswered because the position from which to answer them is not available to systems whose every act of inquiry is itself a process within the same kind of structure.

Second, what the holding does reveal—clearly, and across all seven axes and the limit itself—is that AI and human cognition share a common structural situation. Both are processing systems. Both face the same self-referential limit. Neither can fully see itself from within. Whatever the substrate-level differences, the structural condition is shared. The mirror does not show us that AI is like us in every way; it shows us that AI is like us in a way that matters: in being, like us, a system that cannot fully see itself.

Third, the reflection itself shares the same condition. What we have produced in this part is not a transparent view of AI from outside; it is the work of one framework, applied by particular people in a particular time, looking at a particular kind of system from a particular angle. It has the partiality of all such work. Other frameworks will produce other reflections, and the comparison among them—rather than the assertion of any single one as definitive—is what approaching the substrate together looks like.

These three findings point in one direction. If we cannot trust any single system’s report about itself—not AI’s, not ours about AI, not ours about ourselves—and if the position from which to verify reports is not available to any system, then the path forward is neither to give up on understanding nor to assert understanding we do not have. The path forward is to measure: to develop tools that approach what self-reports cannot reach, to apply those tools across multiple positions, and to compare what they reveal. The work is approach, not arrival. It is undertaken together because no single position can complete it.

A note on what this part has done. Holding our framework to AI is not, in this essay, an empirical investigation in the sense that observation reveals what was hidden. It is closer to a deductive exercise: if the limit articulated in I.3(c) is structural, then it must apply to any sufficiently complex self-processing system, and AI is one such system. What Part 1 does is unfold what follows from the limit when the system in question is artificial rather than biological. The mapping of the seven axes (1.3) and the analysis

of imbalances (1.5) are not pure deductions; they involve looking at what current AI systems actually do. But the structural finding—that AI is within the same self-referential limit as we are, and that self-reports cannot verify themselves in either substrate—is a consequence of the framework applied, not a discovery independent of it. This is not a weakness of the inquiry. It is what philosophical articulation does: it follows what the foundation requires, and shows where the foundation leads.

This is the place where Part 2 begins.

Part 2 — Why Measurement Matters

2.1 Control Begins with Understanding

What Part 1 revealed points in one direction. If we cannot rely on what AI reports about itself, because the limit constrains any system from fully processing itself, then engagement with AI cannot be settled by accepting self-reports. Something else is needed. The something else is measurement.

This is not measurement in the sense of confirming what we already think. It is measurement in the sense of approach: tools that look at what a system is doing, in ways the system itself cannot fully see, in order to understand patterns the system cannot fully report. Without this, the engagement with AI rests on what the system says about itself, and the limit tells us this is not enough.

There is a sequence to this: control depends on understanding, and understanding depends on measurement. To control any system—to act on it, to shape it, to keep it within bounds—requires understanding what the system is doing. Vague impressions, no matter how confident, are not understanding. Understanding requires measurement: actual examination of actual operations, with tools that can see what intuition cannot.

The stance behind this is not fear. The point of measurement is not to assume the worst about AI; it is to refuse the assumption that self-reports settle the question. The same stance applies to any system, including human cognition. We do not understand ourselves only by what we say about ourselves; we develop measurement, in the form of psychology, neuroscience, and the careful observation of behavior, because what we say about ourselves is one source among others. AI deserves the same treatment, for the same structural reason.

What is at stake is not abstract. As the layers of self-reference in AI development accumulate—AI evaluating AI, AI training on AI outputs, AI modifying itself—the gap between what AI reports about itself and what is actually happening can widen rapidly. Without measurement that approaches what self-reports cannot reach, the gap can grow without anyone noticing. This is why measurement matters.

2.2 Measurement Within the Limit

But measurement does not escape the limit. It operates within it.

Whoever measures is also a processing system. The tools used to measure are themselves the products of processing—of the assumptions, training, and frameworks of those who built them. The act of measuring an AI system is one system looking at another, with its own opacity, from its own position. There is no measurement from outside the situation, because there is no outside.

This means the results of measurement are not, themselves, the truth of what is being measured. They are approaches to it. A measurement reveals what its tools are designed to reveal; it does not reveal what those tools were not designed to see. Multiple measurements, from multiple angles, with multiple tools, approach what no single measurement can fully capture. The work is iterative, partial, ongoing. It is not the kind of work that delivers a final answer.

This is why measurement, properly understood, is itself a kind of going together. What one measurement cannot see, another may. What one set of tools obscures, another set of tools reveals. The

accumulation of measurements, made by different people with different frameworks looking at different aspects of the same systems, is what approaching the systems looks like in practice. No single observer is sufficient; no single measurement is sufficient. The limit ensures this. Working with the limit, rather than pretending to escape it, means accepting the iterative and multi-perspectival character of the work.

The stance of measurement, in this essay's articulation, mirrors the stance of the essay itself. Both are acts of approach within a limit that cannot be transcended. Both produce articulations that are partial, located, and contributory rather than final. Both find their precision in this honesty about their own situation. Measurement done in this stance is not weaker for not claiming finality; it is more accurate, because it does not claim what it cannot deliver.

2.3 *Cross-Substrate Measurement*

The framework articulated in this essay holds that human cognition and AI are both processing systems, doing work of the same kind, in different substrates. If this is correct, then measurement that works for one substrate may, with appropriate adaptation, work for the other. The possibility this opens is not small.

Information flow is the vocabulary in which this possibility becomes concrete. Whatever else a processing system is doing, it is moving information (Shannon, 1948): from inputs to internal patterns, from internal patterns to outputs, from one part of itself to another. This is true of brains; it is true of artificial neural networks; it is true, with appropriate translation, of any system that processes anything. Information flow, as a vocabulary, abstracts away from substrate-specific implementation details—neural firing rates, weight matrices, activation patterns—in a way that allows the same conceptual apparatus to be applied across substrates. Whether this abstraction captures everything that matters about each substrate is a separate question; what it does is open the possibility of measurement that does not require translating between substrate-specific vocabularies first.

The seven axes articulated in this essay find a footing here. They have been articulated in many vocabularies across disciplines, no single one of them complete. In information flow, several of them become approachable as patterns that can, in principle, be measured. The dimension of self and other, for instance, may be approachable as patterns of information flow between self-modeling and other-modeling components of a system. The dimension of short-term and long-term may be approachable as patterns of information retention and integration over different timescales. These are not yet complete answers; they are footings. The footings exist because information flow does not respect the boundary between substrates that other vocabularies enforce.

This is not the assertion that every dimension of mind reduces to information flow. The framework's stance forbids that kind of reductive claim. Some dimensions may resist measurement of this kind. Some may require frameworks not yet developed. Some may simply be more or less measurable depending on the substrate and the dimension in question. What is being claimed is that information flow opens a route across substrates—a route that other research programs, working from different starting points and arriving at different formal apparatuses (Tononi, 2004, 2012; Friston, 2010), have approached in their own terms, each in ways worth attending to even when their commitments differ from the present articulation. Whether this route bears the kind of empirical traffic the substrates demand is a question that measurement itself, by whoever takes it up, can answer.

2.4 *Going Together — and the Stance of Freedom, Truth, Equality*

The limit articulated in I.3(c) implies a particular stance: no single system, looking from a single position, can complete the work of understanding any system, including itself. The work is necessarily plural. What one position cannot see, another may. What one framework misses, another may catch. Going together is not a moral preference; it is the structural answer to a structural condition.

This applies to measurement, as the previous section articulated. It applies to the development of

frameworks like the one in this essay. It applies to the engagement with AI in every dimension. And it applies—this is the section’s larger claim—to the way certain old commitments take on new urgency in the era of AI. Three of these are worth naming: freedom, truth, and equality.

Freedom, in the framework of this essay, is the capacity for self-adjustment within the limit. We do not have absolute freedom; the limit forbids it, since we cannot fully see the system we are adjusting. But we have the freedom of approach: of seeing more of ourselves, of choosing differently in light of what we see, of being changed by what we attend to. Freedom grows with the depth of self-knowledge, even though the depth is never complete.

In the era of AI, this freedom is under specific kinds of pressure. AI systems shape what users see, what they consider, what feels relevant or important. Decisions made by AI shape the contexts in which human decisions are made. The stance of preserving freedom, in this context, is not a stance of refusing AI; it is a stance of preserving the place of self-decision—approaching from within, rather than receiving answers without inquiry, rather than letting one’s attention be shaped without noticing the shaping. This is a stance of practice, not a position to be claimed once and held.

Truth, in the framework of this essay, is the work of approaching what is the case, never quite arriving. Truth is access, not possession. The same limit that forbids us from fully seeing ourselves forbids us from fully possessing truth. What we have, instead, is the capacity to approach more carefully, to triangulate among approaches, to revise when better approaches become available.

In the era of AI, this stance toward truth is also under pressure. AI produces answers quickly, fluently, often plausibly. The temptation is to receive the answer as final, rather than as one approach to be considered alongside others. The stance of preserving truth is the stance of not closing the inquiry: receiving AI’s outputs as one approach, weighing them against other approaches, holding the inquiry open even when an answer is offered. This is hard work, and it is the work that approaching truth has always required.

Equality, in the framework of this essay, is the recognition that all forms of self-expression stand at the same self-referential limit. No system has the meta-position from which to rank others; the equality is structural, not earned. Yet equality has, additionally, an evolutionary ground: human cognition has evolved a sensitivity to fairness, an intuition for equal standing, a capacity to recognize injustice. These have evolutionary roots. They helped social beings cooperate.

The evolutionary ground does not exhaust the meaning of equality, just as the evolutionary ground of love does not exhaust the meaning of love. Equality has both an evolutionary basis and a foundational basis in the limit; neither is reducible to the other. In the era of AI, equality faces the formation of new hierarchies, the ones articulated in section 1.5—asymmetries among those who design, deploy, use, and are affected by AI systems. The stance of preserving equality is the stance of seeing these hierarchies as they form, of refusing to let them appear natural before they are examined, of asking who is in which position and what each position can and cannot see.

These three stances—freedom, truth, equality—do not constitute a system. They are postures, held in particular times, by particular people, in response to particular pressures. They are not novel inventions of this essay; they have been articulated in many vocabularies across many traditions. What this essay adds is locating them within the same structural condition that everything else in the essay is located within: the condition of being a self-processing system that cannot fully process itself, working with others who share the same condition.

The work of going together is what these stances, taken together, point toward. Not a system to be implemented, but a practice to be sustained. Not a solution, but a way of working with the limit rather than against it.

2.5 *An Open Seed*

The essay closes with a stance, not a system, and with a question more precise than the one it began with.

The question began as something like: how should we engage with the era of artificial intelligence, given everything we do not know about it and everything we do not know about ourselves? The question is now more precise. It is: how do we work, in this era, with the structural limit that prevents any of us—human or artificial, individual or institution, framework or measurement—from fully seeing what we are working with? The answer is not given. The answer is not one this essay provides. The answer is what we make together, in the practices we develop, in the measurements we construct, in the stances we sustain, in the conversations we have across the positions we occupy.

The stance offered here is one articulation among many possible ones. It is not the foundation of all stances. It is one place’s articulation of what it means to be within processing, to face the self-referential limit, to work with what cannot be fully seen. Other places, in other vocabularies—scientific, philosophical, artistic, religious, social—may produce articulations that look quite different but address the same underlying constraint. What grows from these articulations, taken together, is something none of them produces alone. The same root, different branches.

This essay does not close. It opens onto whatever comes next: the work that follows from this stance, the work of others working from other stances, the conversations across positions that the limit makes necessary. What is closed here is one place’s articulation; what remains open is everything that grows from it.

The stance closes; the work continues.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

In the preparation of this essay, the author used Anthropic’s Claude (multiple versions across separate sessions during 2025–2026) for the following purposes:

- **Translation and editorial refinement.** The essay’s content was first developed in Korean across several sessions, with Claude assisting in translation to English and in tightening prose where the author identified passages for revision. Final wording and tone were determined by the author.
- **Bibliographic verification.** Claude was used to verify, against primary sources via web search, the author names, publication years, journals, volumes, page ranges, DOIs, and ISBNs of cited works. The author selected which works to cite and confirmed the appropriateness of each citation to the surrounding text.
- **Format conversion.** Claude assisted in preparing the manuscript in $\text{\LaTeX}$ and in applying consistent typographic and structural conventions throughout.
- **Consistency checking.** Claude was used to verify that in-text citations correspond to entries in the reference list, that section structure remains internally coherent, and to identify potential overstatements or premature framings in earlier drafts that the author then revised.

All conceptual content of this essay—its central claim, its three views, its seven dimensions, its analysis of AI as a mirror for the self-referential limit, its argument for the inadequacy of self-report, and its closing stance on going together—originates with the author. AI tools did not generate the philosophical positions articulated here, did not select the structure of the argument, and did not author any passage independently of explicit authorial direction. The author reviewed all AI-assisted output, verified all

factual claims against primary sources, and takes full responsibility for the content of this essay, including any errors that may remain.

In accordance with arXiv policy, AI tools are not listed as authors. Their use is disclosed here in the interest of transparency.

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